

Afshad Y. Sidhwa

Karachi, Pakistan | (+92) 348-9098298 | afshadsidhwa@gmail.com | [LinkedIn](#): afshadsidhwa | [GitHub](#): afshadGit

EDUCATION

Habib University

B.Sc. in Computer Science (6/8 semesters complete)

Karachi, Pakistan

Expected graduation: May 2027

Karachi Grammar School

A Levels: Computer Science, Mathematics, Economics

Karachi, Pakistan

May 2023

EXPERIENCE

Undergraduate Research Assistant — PPAGE (Open-Source MARL Research Engineering), Habib University

Karachi, Pakistan

June 2026 – Present

- Top contributor to **PPAGE**, an open-source, PyPI-packaged multi-agent RL research testbed (Apache-2.0); designed the modular algorithm/environment architecture, YAML-based experiment configs, and CSV logging/experiment-runner infrastructure.
- Implemented **PyTorch** MARL baselines from scratch — DQN (Double/Dueling), Actor-Critic, A2C, A3C (async “Hogwild”) — plus a **MixedTrainer** for per-team algorithm asymmetry; designed **JAL-GT**, a correlated-Q baseline solving a per-timestep correlated-equilibrium LP.
- Diagnosed a shared ~5% capture-rate failure across two algorithms to an observation-encoding bug rather than misattributing it to either baseline; ran controlled multi-seed experiments (**JAL-GT**: **+1.8pp** capture rate vs. CQL at ~70× compute cost) and grew test coverage to ~81%; codebase now being scoped for AAMAS/EAAI and JOSS submission.

Web Development Intern — VSI Cloud App Development & Maintenance, Systems Limited

Karachi, Pakistan

June – July 2025

- Designed and built **SkillBoard**, a full-stack workforce-intelligence platform for skill-based task allocation, reducing burnout risk through real-time workload and capacity tracking.
- Built the backend with FastAPI/Python and MS SQL Server to manage tasks, employee-skill mappings, and dynamic workloads; redesigned the allocation logic from role-based to skill-based.

Technology & Business Intern, Bank Al Habib Ltd.

Karachi, Pakistan

July 2022

- Rotated through Database Systems, Marketing, and Software Engineering; gained enterprise SQL and banking-application exposure.

PROJECTS

Hybrid Genetic Algorithm for Multi-Agent Route Optimisation

Python, Genetic Algorithms, Metaheuristics

- Built a 5-component hybrid GA (path-guided initialisation, crossover, tabu mutation, local search) achieving 13.6–23.4% cost reduction over baselines on six TSPLIB instances.
- Applied to Pakistani fuel logistics (142 stations) via multi-brand hub modelling across Shell, PSO, and Total/PARCO fleets, achieving ~20% improvement through inter-route optimisation.

Automated Hair Type Classification & Product Recommendation System

Python, PyTorch, OpenCV

- Built an end-to-end computer-vision pipeline (ResNet-18 classification, OpenCV colour/volume estimation) deployed via Streamlit with personalised recommendations; explored YOLOv8 segmentation and geometric metrics toward porosity assessment.

TECHNICAL SKILLS

Programming: Python, C++, SQL, JavaScript, HTML/CSS, Verilog, LaTeX

Web & ML Frameworks: React, FastAPI, PyTorch, Scikit-learn, XGBoost, OpenCV, Streamlit, Tailwind CSS

AI & Algorithms: Reinforcement Learning (DQN, Actor-Critic, MARL), Genetic Algorithms, Metaheuristic Optimisation

Tools & Databases: Git/GitHub, Linux, MS SQL Server, MySQL, SQLite3, Vercel

VOLUNTEER

Fundraising Performer & Volunteer, Harmony in Helping Hands (piano ensemble)

Karachi, Pakistan

2016 – Present

- Performed in 8 large-scale fundraising concerts for charitable organisations, including Indus Hospital Children's Cancer Ward, Chitral Education Fund, ABSA, BMH Parsi General Hospital, National Institute of Child Health, Lady Dufferin Hospital, and SOS Village